Gruppo FS

The Mobility Leader

Mobility analysis for exceptional events

38th America's Cup in Naples, using Big Data

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POLITECNICO MILANO 1863

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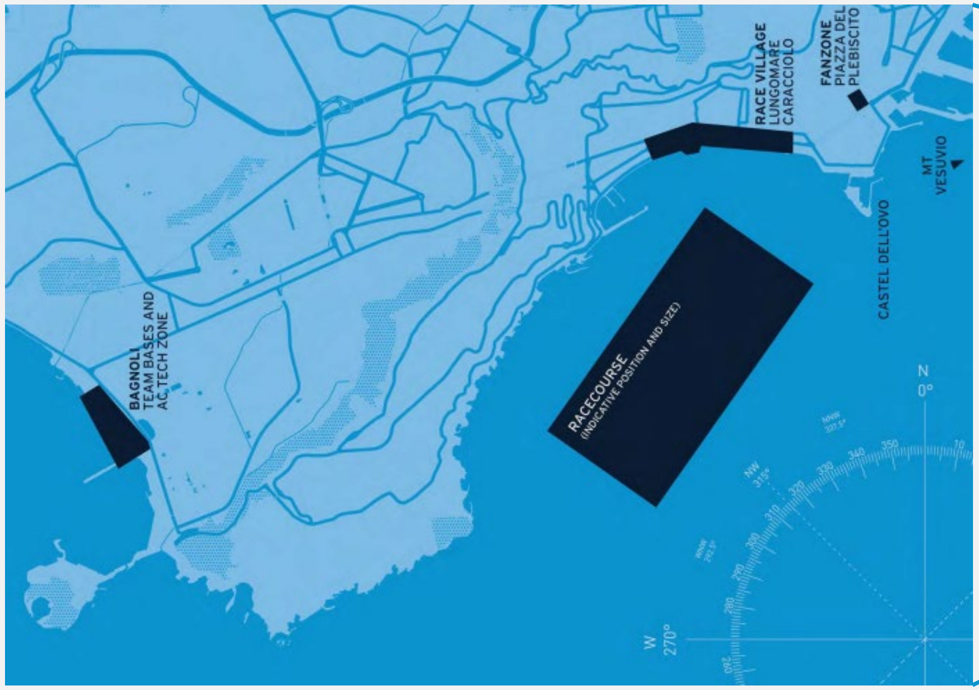
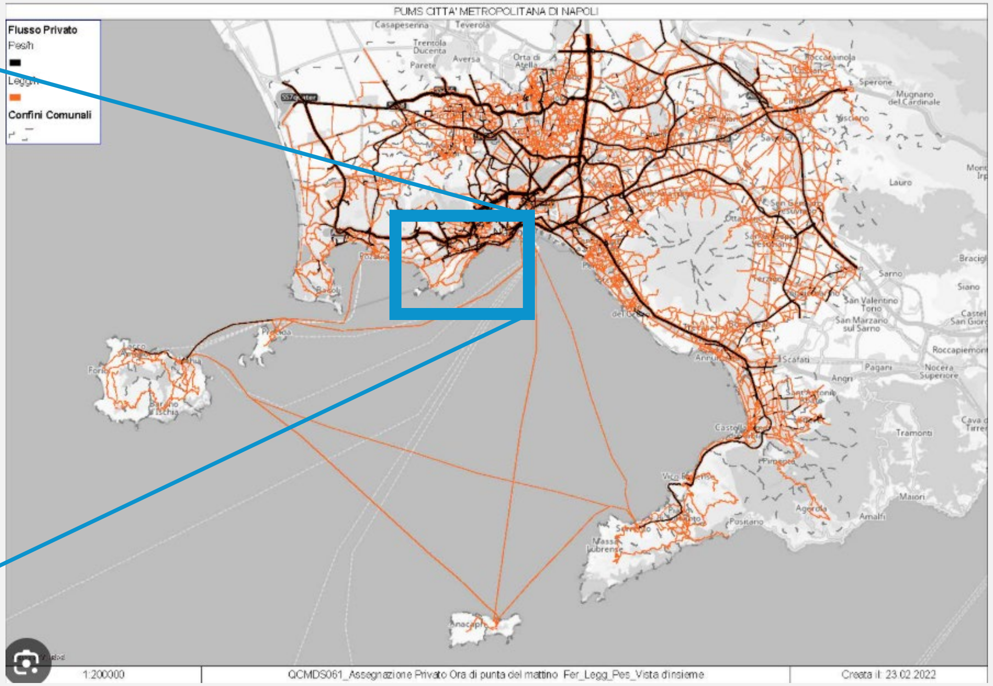
Event and Location

*"The **Louis Vuitton 38th America's Cup** will take place in **spring** and **summer 2027** on the beautiful bay of Naples, Italy under the watchful shadow of Mt Vesuvius. For the first time ever the Louis Vuitton Cup and Louis Vuitton America's Cup Match will be sailed in Italy, a country with one of the most colourful and enthusiastic America's Cup histories."* (from the event press kit)



Picture from America's Cup website



Picture from the Sustainable Urban Mobility Plan of the Metropolitan City of Naples

Proposed methodology

Step 1 : "As is" mobility

- **O-D matrices** from MND
- **Preliminary data analysis**: anomalies **detection**, statistical **testing**
- **Geospatial** analysis and mapping, **chart** visualization

Step 2 : Demand forecast

- Retrieve **data from previous editions** regarding the number of visitors
- Identify **accommodation facilities**
- Estimate **new flows** toward the event's points of interest
- Possible flows toward **other points of interest** in the area

Step 3 : Services improvement

- Analysis of the **existing public transport** offer
- Identification of **new infrastructures** under construction and **future services**
- Services **enhancement hypotheses** based on the estimated increase in mobility demand

The source of information

Mobile phone users

The devices interact with the network many times per minute

Antennas

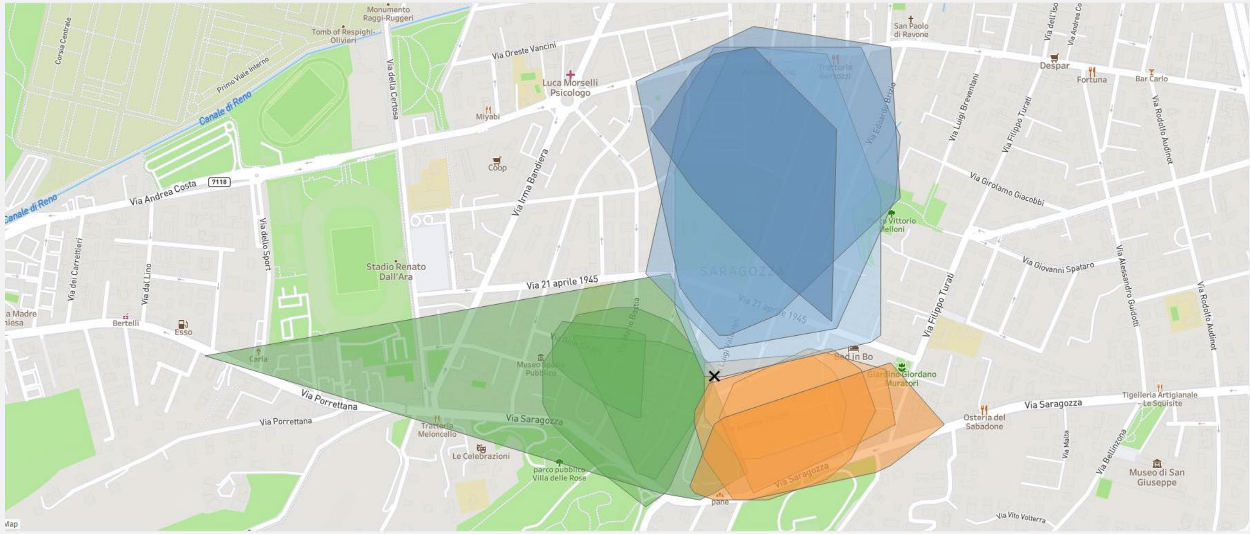
They are the access point to the network for mobile phones

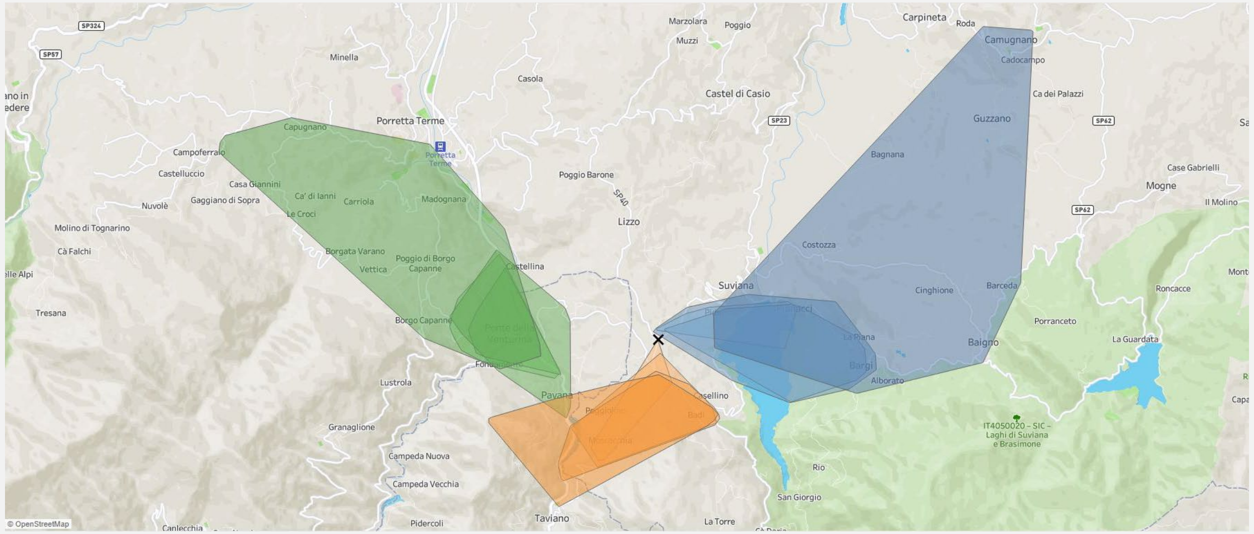
Coverage area

Range

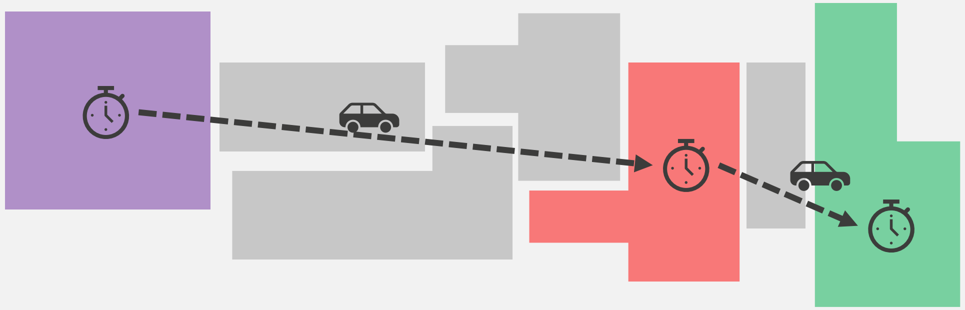
Shape

Obstacles





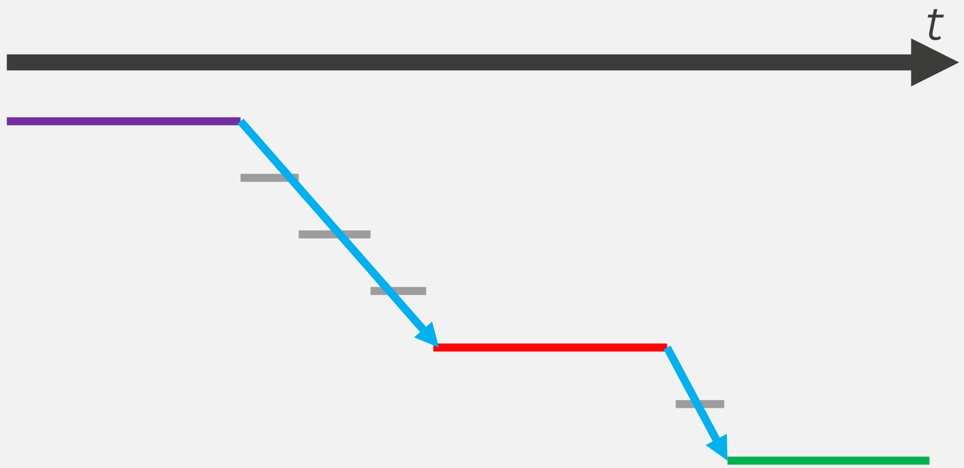
O-D matrices methodology: zones and thresholds



Discretization of the territory based on zones.

Time threshold

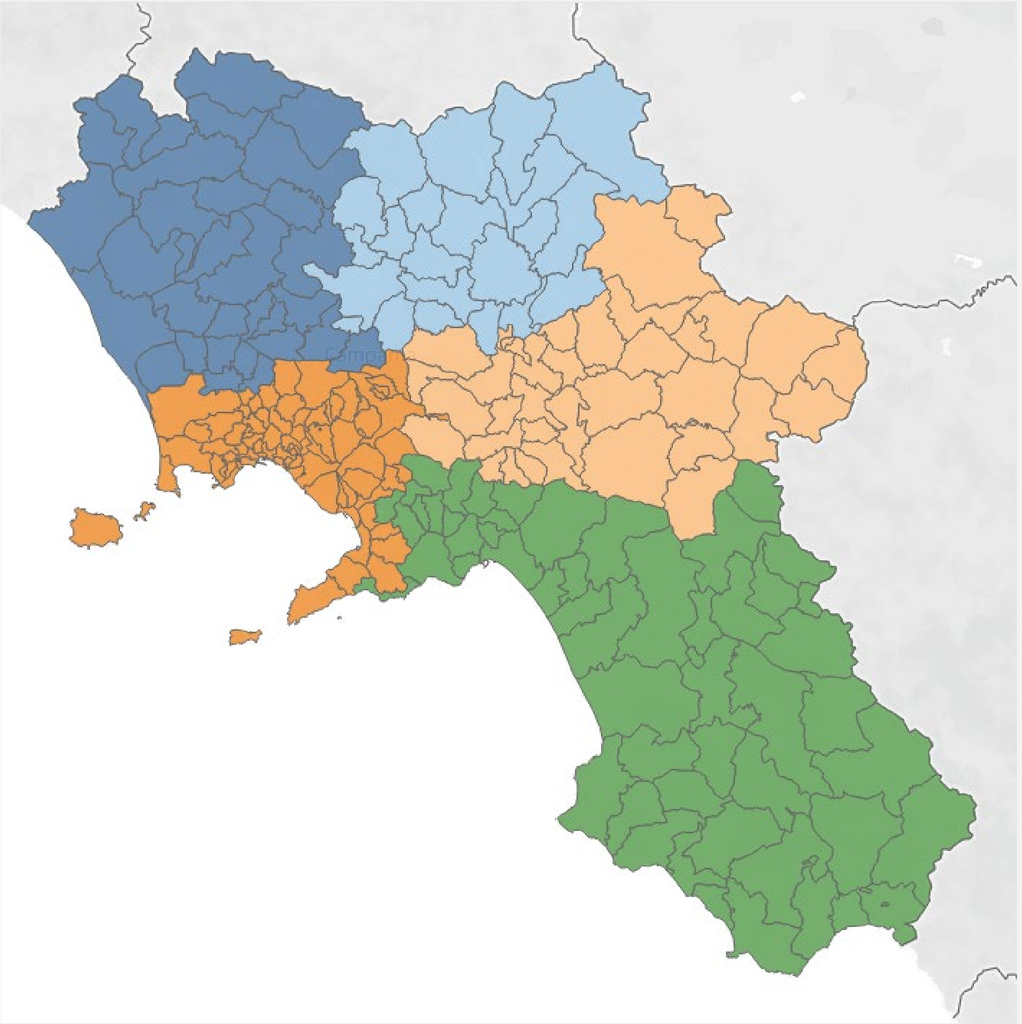
Origin and Destination



The zones pattern and the time thresholds are crucial to identify mobility behaviours, and they affect heavily the results.

Relationship between areas shape and extension and the time requested for crossing that area.

Study datasets



- **222 zones** in Campania region, 77 zones in Napoli province
- **1 hour** stop threshold
- **Number of trips** from zone A to zone B **where A <> B**
- **Further information** such as: time slot, week day, users residence class, users nationality, transport mode
- **Hidden data** ("mascherato") because of GDPR constraints
- **Two datasets** with different details:
  - Main dataset: **sistematicity** and **nationality**
  - Supplementary dataset: **transportation mode**
- **Files provided**: two datasets, shapefile, datasets description, lookup tables

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Thank you

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